HE A 776

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Extracted Text

What the doctor thus calculates, from deposition, with regard to mandind, he actually experimented op-on horfes, dogs, fallow-does, cz, by fixing tubes, in<y_0.1734><class_Text>

<x_0.5459><y_0.0625>cerifices opened in their veins and arteries ; by obfer-ving into federal heights, or which the blood rode in these tubes, as they lay on the ground, y and by mea- firing the capacities of the ventricles of the heart, and orifice of the arteries. And, that the reader may the more readily compare the faid eliminates together, he has given a table of them, ranged in the following order.

\begin{tabular}{ccccccccccc} **The feeral animals.** & **Weight of each.** & **Height of the blood in the tube from the gui- urine.** & **Capacity of the left ventricle of the heart.** & **Area of the orifice of the aorta.** & **Velocity of the blood in the sorta.** & **Welocity of the blood in the sorta.** & **Flows.** & **How much in a minute** & **Weight of the blood fur- flined by the left ventricle contracting.** & **Area of transferfe fiction of defending aorta.** & **Area of the transferfe fiction of defending aorta.**\\ & Pounds, Ounces. & Inches. & Retelaches. & Cubicinches & Square inches. & Pete andiae circa in minute. & Minutes. & Pounds. & Pounds. & No of pullies in a minute. & Square inches.\\ Man & 160 & On draining. & 7 6 & 1.659 & 3.318 & 0.4187 & 56.55 & 34.18 & 34.38 & 51.5 & \\ Horfe 1th. 2d. 3d. & 12 5 & 9 8 & 8 9 & 10 2.5 & 1.036 & 86.85 & 60 88 & 13.75 & 113.22 & 86 38 & 0.369\\ Ox & 835 1600 & 12 5 & 9 6 & 10 1.539 & 86.95 & 76.95 & 18.14 & 113.22 & 86 38 & 0.677 0.912 & 0.84\\ Sheep Doe & 91 & 5 & 9 6 & 1.85 & 0.172 0.476 & 174.5 & 20 & 4.593 & 36.56 & 50 0.984 & 0.07 0.012\\ Dogs ift. & 52 24 & 5 6 & 6 8 8 & 1.172 & 0.196 & 144.77 & 11.9 & 4.34 & 33.61 & 97 & 0.106 0.034\\ 2d. 3d. 3d. & 18 5 & 5 7 & 8 8 & 0.185 & 0.118 & 130 & 7.8 & 2.3 & 19.8 & 11.1 & 0.07 0.022 0.009\\ 4th. & 12 8 & 4 & 8 3 & 0.633 & 0.101 & 120 & 6.7 & 1.85 & 11.1 & 0.061 & 0.015 0.007\\ \end{tabular}

tery of a man, and that the inward area of the left ventricle of his heart is equal to fifteen figures under; their multiplied inn =+1 for, give 13 to cubicinches of blood, which prefix on that ventricle, when it drift begins to control, a weight equal to 15 y.g pounds.

Matched Wikipedia Article

Title: Inward investment
Page ID: 347261
Revision ID: 1352223370
URL: `https://en.wikipedia.org/wiki/Inward_investment`

Source record

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tery of a man, and that the inward area of the left ventricle of his heart is equal to fifteen ſquare inches; theſe multiplied into 7 1/4 feet, give 1350 cubic inches of blood, which preſſes on that ventricle, when it firſt begins to contract, a weight equal to 15.5 pounds.

What the doctor thus calculates, from ſuppoſition, with regard to mankind, he actually experimented upon horſes, dogs, fallow-does, &c. by fixing tubes, in orifices opened in their veins and arteries; by obſerving the ſeveral heights, to which the blood roſe in theſe tubes, as they lay on the ground; and by meaſuring the capacities of the ventricles of the heart, and orifices of the arteries. And, that the reader may the more readily compare the ſaid eſtimates together, he has given a table of them, ranged in the following order.

The ſeveral animals.	Weight of each.	Height of the blood in the tube from the jugular vein.	Height of the blood in tubes fixed to arteries.	Capacity of the left ventricle of the heart.	Area of the orifice of the aorta.	Velocity of the blood in the aorta.	Quantities of blood equal to the weight of the animal, in what time.	How much in a minute	Weight of the blood ſuſtained by the left ventricle contracting.	N° of pulſes in a minute.	Area of tranſverſe ſection of deſcending aorta.	Area of the tranſverſe ſection of aſcending aorta.
	Pounds. Ounces.	Inches.	Feet Inches.	Cubic inches.	Square inches.	Feet and inches in a minute.	Minutes.	Pounds.	Pounds.		Square inches.	Square inches.
Man	160	On ſtraining.	7 6	1.659	0.4187	56.55	34.18	4.38	51.5	75		
				3.318		113.3	17.5	9.36				
Horſe 1ſt.			8 3									
2d.			9 8									
3d.	825	12 52	9 6	10	1.036	86.85	60	13.75	113.22	86	0.677	0.369
Ox	1600			12.5	1.539	76.95	88	18.14		38	0.912	0.84
												right. left.
Sheep	91	5 1/2 9	6 5 1/2	1.85	0.172	174.5	20	4.593	36.56	65	0.094	0.07 0.012
Doe			4 2	9	0.476						0.383	0.246
												right. left.
Dogs 1ſt.	52	0 6	6 8	1.172	0.196	144.77	11.9	4 34	33.61	97	0.106	0.041 0.034
2d.	24	5 7	2 8	1	0 185	130.9	6.48	3.7			0.102	0.031 0.009
3d.	18	5	4 8	0.633	0.118	130	7.8	2.3	19.8		0.07	0.022 0.009
4th.	12 8	4	3 3	0 5	0.101	120	6.7	1.85	11.1		0.061	0.015 0.007

HEAT, in phyſiology, one of the ſecondary qualities of bodies, produced by fire, and oppoſed to cold.

Under the article fire, we conſidered the ſun as the principal ſource of heat upon the earth's ſurface, and the confines of the earth and atmoſphere: without this, all the bodies upon our globe would doubtleſs grow rigid, lifeleſs, and fixed. It is this that ſtirs within them, as the main ſpring of their actions. Hence vegetation and animalization are evidently promoted; and hence the ocean and the atmoſphere continue in a fluid ſtate.

Heat in us is properly a ſenſation, excited by the action of fire; or it is the effect of fire on our organs of feeling. Hence it follows, that what we call heat is a particular idea or modification of our own mind, and not any thing exiſting in that form in the body that occaſions it. Heat, ſays Mr Locke, is no more in the fire that burns the finger, than pain is in the needle that pricks it. In effect, heat in the body that gives it, is only motion; and in the mind, only a particular idea.

Heat in the hot body, according to 'S Graveſande, is an agitation of the parts of the body, made by means of the fire contained in it: by ſuch an agitation a motion is produced in our bodies, which excites the idea of heat in our mind; ſo that heat in reſpect of us is nothing but that idea, and in the hot body nothing but motion. If ſuch motion expel the fire in right lines, it gives us the idea of light; if in a various and irregular motion, only heat.

Heat, with reſpect to our ſenſations, or the effect produced on us by a hot body, is eſtimated by its relation to the organ of feeling; no object appearing to be hot, unleſs its heat exceed that of our body. Whence the ſame thing to different perſons, or at different times to the ſame perſon, ſhall appear both hot and cold. The degree of heat is meaſured by the expanſion of the air, or ſpirit in the thermometer.

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Figure 1: *
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